Q1. Fill in the blanks for the given linear optimization model.

Maximize	$Z = 20X_1 + 50X_2$
Subject to	$1.2X_1 + 4X_2 <= 240$
	$0.5X_1 + X_2 <= 81$
	$X_2 <= 40$
	$X_1, X_2 >= 0$

```
# import the cvxpy package
Import _____________

# decision variables
X1 = _____________

X2 = _____________

# objective functions
objective = ___________________

# constraints
constraints = [
    ____________________________,   # assembly
    ____________________________,   # inspection
    ____________________________,   # chips
    ____________________________,   # non-negativity
    ____________________________    # non-negativity
]

# Formulate the problem
prob = ____________________________

# Solve the problem
____________________

# Print the results
print("Optimal solution of X1:", ____________________________)
print("Optimal solution of X2:", ____________________________)
print("Optimal value:", ____________________________)
print("Optimization status:", ____________________________)
```

Q2. Write the Python codes to solve the given linear optimization model.

Minimize $Z = 6000X_1 + 1000X_2$

Subject to $300X_1 + 20X_2 >= 66$

$30X_1 + 10X_2 >= 9$

$X_1, X_2 >= 0$

```
# import the cvxpy package


# decision variables



# objective functions

# constraints




# Formulate the problem


# Solve the problem


# Print the results
```

Q3. Setup the linear optimization model to solve the following problem and then write the Python codes to solve the model.

K Electronics plans to hire skilled workers and apprentices as it expands its business operations. The cost of employing a skilled worker is 10,000 won per hour, while an apprentice[1] costs 5,000 won per hour. The company has a budget of 150 million won for this hiring initiative, and under the terms of the labor union agreement, the entire amount must be allocated to this plan. Taking current operational conditions into account, K Electronics intends to limit the employment of skilled workers to a maximum of 10,000 hours. Additionally, the labor union requires that apprentices be employed for a minimum of 12,000 hours. Current cost analysis indicates that skilled workers generate a profit of 3,000 won per hour, whereas apprentices generate 1,250 won per hour. Management aims to formulate a hiring plan that maximizes total profit.

[1] apprentice: a person who is learning a work from a skilled employer

< Linear optimization model>

```
< Python code>
```